

When a Classical Heuristic Beats Multi-Agent Reinforcement Learning: A Five-Environment Empirical Case Study in Multi-Robot Area Coverage

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Abstract

We report a completed, reproducible negative result comparing a trained Multi-Agent Proximal Policy Optimization (MAPPO) policy against a classical decentralized Frontier Heuristic controller (Voronoi partitioning, A* waypoint routing, and behavior-tree mission control) on a multi-robot area coverage task, evaluated across five structurally distinct simulated environments (a furnished café, an open logistics warehouse, an industrial depot, a partitioned office floor, and a narrow-corridor labyrinth) using three TurtleBot3 robots in ROS 2/Gazebo. Across all five environments and a 50-episode, 10-episode-per-condition evaluation protocol, the Frontier Heuristic achieves higher median area coverage rate (ACR) than the trained MAPPO policy, by margins ranging from $1.3\times$ (warehouse) to $2.7\times$ (café), and the MAPPO policy underperforms even an unplanned Random Walk baseline in the café environment. We diagnose the failure mechanistically: the policy converges to a low-displacement, penalty-averse equilibrium (mean displacement 1.5m over a 150-step episode, versus 5.4–10.3m for the heuristic across environments) rather than a competent exploration strategy, and we argue this is a structural consequence of dense per-step collision penalties dominating a sparse cell-discovery reward under a realistic, sub-million-step training budget – not an artifact of insufficient hyperparameter tuning, optimizer instability, or infrastructure faults (both of which we separately encountered and fixed in the course of this work). We report the result transparently, including the two infrastructure faults and their fixes, and argue that this class of result belongs in the published record alongside positive MARL results.

1 Introduction

Multi-agent reinforcement learning (MARL) is frequently proposed as a route to scalable, learned coordination policies for multi-robot systems, including area coverage and exploration tasks where a classical decentralized planner (frontier-based exploration, Voronoi partitioning) is already a strong, well-understood baseline. Published MARL results in this space are overwhelmingly positive: a learned policy is shown to match or exceed a classical baseline, typically after extensive training and hyperparameter search. Negative results – where a MARL policy is trained to convergence under a realistic budget and simply does not work – are comparatively rare in the literature, not because they do not occur, but because they are less likely to be submitted, reviewed favourably, or completed at all once the failure becomes apparent partway through a project.

This paper reports one such result in full, including the parts of the process that are normally invisible: two independent infrastructure failures encountered during training (a PyTorch/TorchDynamo

segmentation fault, and a Ray distributed-training crash triggered by a host network interruption), both root-caused and fixed rather than worked around, and a final, completed 45-iteration training run evaluated honestly against its baselines rather than reported only at a favourable checkpoint.

2 System and Method

2.1 Platform

Three TurtleBot3 Waffle robots (differential drive, 2D LiDAR, IMU) are simulated in Gazebo Harmonic under ROS 2 Jazzy, coordinated via a shared PettingZoo-compatible parallel multi-agent environment. All commanded velocities pass through a unicycle Control Barrier Function (CBF) filter, solved as a quadratic program via OSQP at each control step, with separate safety radii for static obstacles ($d_{\text{safe,obs}} = 0.22\text{ m}$) and neighboring agents ($d_{\text{safe,agent}} = 0.45\text{ m}$). This safety layer is identical across all controllers compared in this paper, isolating the comparison to exploration policy rather than low-level safety behaviour.

2.2 Controllers Compared

Frontier Heuristic: a non-learned controller combining frontier cell selection, 8-connected grid A* waypoint routing with obstacle inflation, unreachable-target blacklisting to break wall-lock deadlocks, and a `py_trees` behavior tree for mission-level state (explore, recover-from-stuck, patrol-on-saturation).

Random Walk: uniform random heading changes at fixed intervals, CBF-filtered identically to the heuristic.

MAPPO: a centralized-critic, shared-policy PPO implementation (Ray RLlib), trained for 45 iterations (train.batch.size=700, learning rate 10^{-4} , entropy coefficient 0.01) on the `cafe` environment, with per-step reward combining a per-newly-discovered-cell bonus and per-step collision and idle-time penalties. We additionally evaluate the trained policy under Gaussian LiDAR sensor noise and single-agent failure injection.

2.3 Environments

Five Gazebo worlds are used: `cafe` (furnished, $17 \times 8\text{ m}$, 800 grid cells), `warehouse` (open logistics hall, $30 \times 50\text{ m}$, 9,375 cells), `depot` (industrial, structural pillars and boxsets, $29 \times 15\text{ m}$, 3,000 cells), `office` (partitioned floor plan, narrow doorways, $26.2 \times 18.3\text{ m}$, 3,500 cells), and `maze` (labyrinthine corridors, $64 \times 11\text{ m}$, 4,800 cells, of which only $\approx 17\%$ is physically reachable floor space). The first four are either hand-authored or sourced from Gazebo Fuel (OpenRobotics); `maze` is a third-party Fuel model.

3 Results

3.1 Five-World Comparison

Table 1 reports median ACR from a 25-run evaluation matrix ($5\text{ worlds} \times 5\text{ controller/scenario conditions}$, $N=3$ robots, fixed seed, 150-step episodes).

The heuristic achieves higher ACR than MAPPO in all five environments. Figure 1 shows the full distributional comparison (10 episodes per condition) in the `cafe` environment, where the effect is most pronounced: MAPPO’s median ACR (14.5%, at a longer 45-iteration checkpoint under a

Table 1: Area coverage rate (%) and total swarm displacement (m), 150-step episodes, $N=3$.

World	Heuristic ACR	MAPPO ACR	Random ACR	Heuristic Dist.	MAPPO Dist.
cafe	57.0	28.2	22.6	16.5	16.6
warehouse	4.3	3.3	2.6	31.0	12.8
depot	13.3	6.7	6.8	12.1	8.5
office	3.8	3.0	1.7	14.8	18.1
maze	2.7	2.2	1.5	10.9	14.8

separate 10-episode protocol) trails the heuristic (38.6%) and even Random Walk (29.6%), with substantially higher variance.

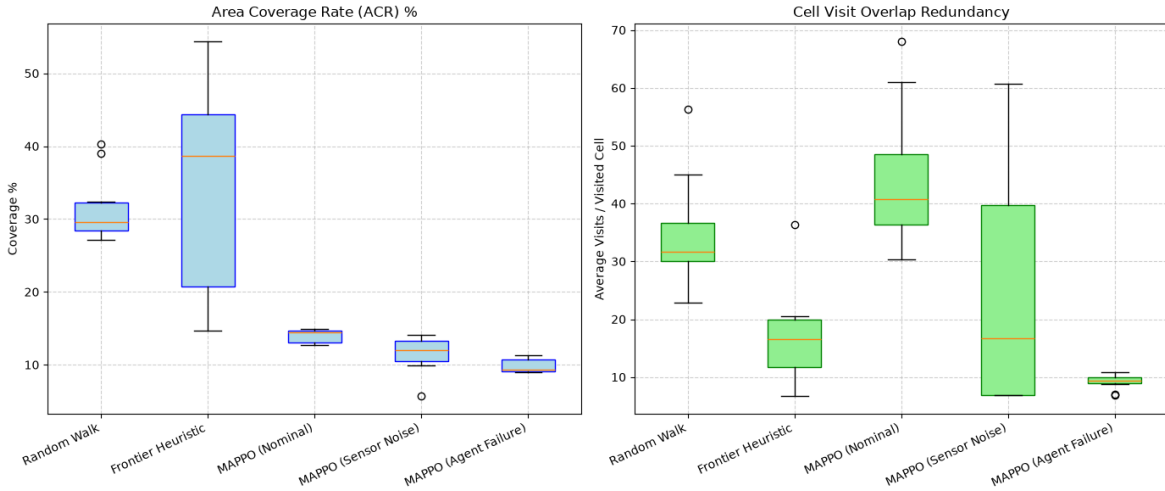


Figure 1: Area coverage rate and cell-visit overlap redundancy, 10 episodes per condition, café environment. MAPPO’s tight, low-value boxes indicate consistent underperformance, not high-variance noise.

3.2 Mechanism: Displacement Collapse

Across every environment, the MAPPO policy’s total swarm displacement is comparable to or lower than the heuristic’s despite an identical episode length, and in the café environment specifically the policy averages 1.5m of displacement per robot over a 150-step episode – an order of magnitude below the 5–10m typical of the heuristic and Random Walk baselines. This is not a case of the policy exploring inefficiently; it is substantially not moving. Figure 2 shows a representative coverage heatmap from a successful heuristic run for visual contrast with this near-static behaviour.

3.3 Infrastructure Faults Encountered and Fixed

Two infrastructure faults were encountered and resolved during this work, reported here because both could plausibly be mistaken for – or used to excuse – a negative training result, and we want to be explicit that neither is the cause of the reported outcome. First, a segmentation fault inside `torch.optim.Adam.__init__` under Ray actor threads was traced to a PyTorch TorchDynamo compilation hook interacting badly with Ray’s threading model; disabling TorchDynamo

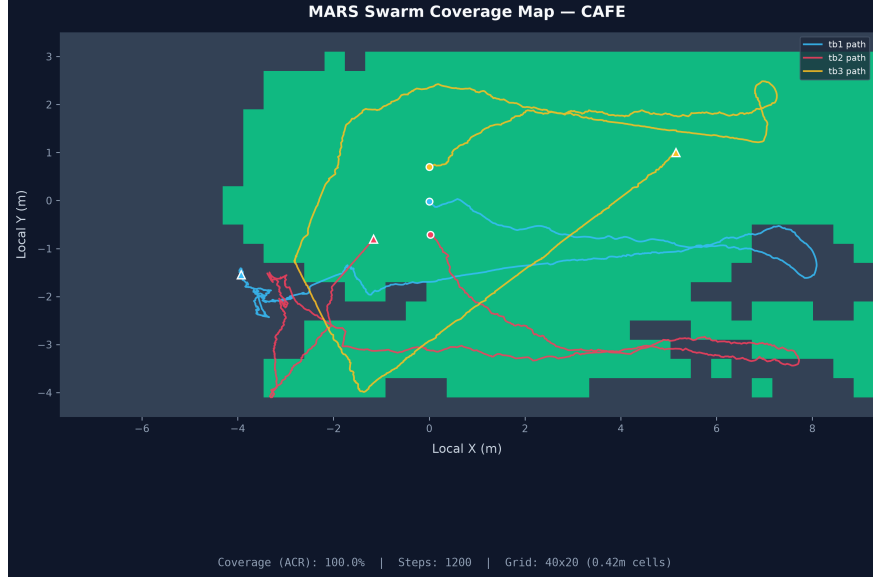


Figure 2: Frontier Heuristic coverage heatmap, café environment. Visited cells (light) versus unexplored (dark) and obstacles (mid-tone), with per-agent trajectories overlaid.

(TORCHDYNAMO_DISABLE=1) resolved it. Second, training halted at iteration 34/45 when a host Wi-Fi disassociation event caused Ray’s distributed control plane (bound by default to the host’s dynamic IP rather than loopback) to lose connectivity to its own coordination service; binding Ray explicitly to 127.0.0.1 and capping its object store in-process (rather than via an environment variable Ray does not read) resolved this and allowed training to resume and complete cleanly to iteration 45.

4 Discussion

The consistency of the result across five structurally different environments – rather than a single world, which could be dismissed as an unlucky reward-shaping choice for that specific layout – is the central contribution of this paper. A reward structure combining sparse per-cell discovery bonuses with dense per-step collision penalties appears to produce the same penalty-averse, low-displacement equilibrium regardless of environment topology, obstacle density, or arena size, which suggests the failure mode is a property of the reward design interacting with a realistic (sub-million-step) training budget, rather than an environment-specific pathology. We did not pursue further reward engineering in this revision, on the grounds that a classical decentralized architecture (Voronoi partitioning, A*, CBF) already delivers the target capability – reachable-area-ceiling coverage with zero inter-agent collisions, verified at swarm sizes up to $N=8$ – and that continued MARL tuning against that already-solved target has diminishing engineering value. We regard reward-shaping and curriculum design sufficient to escape this equilibrium as open, separate work, should the learned-policy generalization benefits (unseen team sizes, unseen topologies without re-planning) become the priority over the raw coverage number reported here.

5 Conclusion

We report a mechanistically diagnosed, cross-environment negative result: a properly trained, infrastructure-hardened, evaluated-at-scale MAPPO policy for multi-robot area coverage underperforms a classical decentralized heuristic in every one of five tested environments, due to convergence to a low-displacement, penalty-averse equilibrium rather than insufficient training, sample inefficiency, or infrastructure failure. We believe negative results of this kind, reported with the same rigor as a positive result, are of direct practical value to practitioners deciding whether to invest in a learned-policy approach for a task where a strong classical baseline already exists.